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Emergence of Player Tactics by expert-guided Machine Learning: An industry tower defence case study☆

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ABSTRACT

Modern games generate a large amount of player data that can enhance the game design and development process. Game developers can potentially utilise data science methods to extract information to inform decision making as they strive to improve the user experience and meet business goals. However, this practice is far from widespread due to the specialist expertise needed. Additionally, games are often complex, where the large number of possible player actions creates datasets with a vast state space and high dimensionality. Additionally, these player actions often require context to fully interpret and analyse. In this emerging research field, a further challenge is in ensuring proposed methods are suitable for commercial game development environments, where genres, available data sources and the production process must be considered.

This work presents the results of an industry–academic collaboration, applying the less common player tactic classification method, individual sequence mining, on a fast paced, commercially available tower defence mobile game. It proposes and evaluates a novel pipeline for validating and discovering player tactics to facilitate game balancing. Rather than being applied to data captured from an analytics framework, the analysis was conducted on data captured from network messages generated by game clients. The real-time nature of these network messages creates potential for this data source to have value beyond tactics classification, with opportunities to integrate into AI pipelines for purposes such as automation.

The resulting mixed methods process demonstrates the ability of using this data source to generate insight on player tactics to game development teams, and it being feasible within the commercial game development process. The pipeline can be applied by other games companies seeking to extract value from data that is collected to make better games for their player base.

1. Introduction

Understanding how players interact with a game can greatly assist game developers. Games can be refined to meet the needs of their players and achieve business goals. In recent years, data analytics has become an important part of this process as game developers seek to gain advantage in competitive markets by better understanding player behaviour [1].

This typically involves telemetry data being logged to an analytics system for later analysis. Data can include player actions, game events, achievements and progress, player retention, monetisation and technical performance data. This data is then displayed to developers via dashboards or reporting systems, and the learnings used for a variety of purposes, including game balance, content creation and monetisation optimisation [2], and understanding different player strategies and tactics.

Off the shelf analytics dashboards typically present surface level analysis relevant for most games, such as player counts, income over time or user acquisition funnels, but do not produce insights required for a deeper understanding of player behaviour, such as patterns of player behaviours specific to a game genre or identifying groups of players on a custom aspect of a game's design. Data science methods such as machine learning e.g. clustering and sequential pattern mining, can be applied to datasets to achieve the desired results [3].

As research into these areas continues, processes and methods that work at scale on real world player data is crucial if these methods are to be adopted in a commercial environment. Additionally, as well as using real world data, there is the need for the process to integrate with commercial game development processes, both technical, such as game engine and data frameworks, and non-technical, such as production or design methodologies.

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In games, data science is commonly applied to player classification [4], where players are grouped according to their characteristics or actions. Typically, this is performed on aggregated data, however aggregation loses information at the level of the player. Retaining player data where there is a large number of possible player actions results in a vast state space and high dimensionality where context is important. The challenges associated with disaggregated data and methods such as sequence mining [5] mean it is less commonly used yet its advantages are clear. Within games, the order of events is often crucial to interpretation. For example, the order in which a player navigates through areas within a game level or the order of execution of moves within a fighting game. Sequence pattern mining retains this crucial context during analysis. Additionally, while most work around player classification uses data captured from analytics systems [1], there are alternatives to be explored, such the data within network messages used in client–server communication.

While there has been some existing research into tower defence games [6–8], there is limited existing research into player tactics classification within the genre, and less within a non-AAA commercial environment.

This paper proposes a new methodology to determine if (a) player tactics can be determined for a fast paced tower defence game using sequence pattern mining (b) network messages are a viable data source for sequence pattern mining, and (c) if a method which integrates domain expertise can operate successfully within a commercial environment.

1.1. Related work

Several data science techniques have been successfully applied to data from games. Clustering is an unsupervised classification technique used to group or cluster data points by their similarity [9]. It can be used as a discovery method, identifying previously unknown groups of data. Within games, it is typically used on gameplay telemetry datasets captured by an analytics framework. This can divide the player base into distinct groups, generating insight into how players are interacting with the game, and how this corresponds to the design intent [10].

Bakkes et al. [11] outlined four main areas of player behaviour modelling, covering modelling actions, modelling tactics, modelling strategies and profiling players. This focussed on approaches that could be incorporated into game AI at runtime. Here, computational efficiency is important due to the constrained resources available for processing, and the need for fast responses. Outside of this environment, free of these constrains, more recent trends in the advancement in machine learning availability and big data methods can be utilised.

Clustering can be applied in various ways. Hou [12] grouped players of an educational game according to their in-game actions, such as social interaction, cooperation and task completion, while Halim et al. [13] used cluster analysis on gameplay data from strategy games to predict players personality type.

While clustering is often applied to aggregated data points generated from gameplay telemetry, it can be combined with sequential pattern mining to group players by sequences of their actions. Sequential pattern mining builds on the data mining method of pattern mining, where associated patterns of data are mined from a dataset [5]. With sequence pattern mining, the time or sequential ordering of the data is important. This method can be applied to a wide range of behaviours and data types, such as looking at patterns of voice input commands [14] or using log files to understand behaviour in gamified applications [15].

Within games, this combination of sequence pattern mining and clustering can provide strong insight into player behaviour, as it takes into account the ordering of events, which can be important context for many games. Kleinman et al. [16] used sequence pattern mining in combination with a previously developed labelling methodology, Interactive Behaviour Analysis, to create a scalable, human in the loop mixed methods approach of examining teamwork within the game DOTA2, a multiplayer real time strategy game. Using data from the game Minecraft, Saadat and Sukthankar [17] used sequence analysis and the concept of motifs to examine player behaviour around activities such as building, fighting and mining, as well as player to player interactions and cooperation.

Some research focusses on real world player data. Drachen et al. [18] applied clustering to a large dataset of player actions from two commercial multiplayer games to identify player types, with a focus on creating actionable groupings of use to developers iterating their game content or aiding decision making. Fu et al. [19] focussed on segmenting players of a MMORPG based on a novel retention metric across a dataset of millions of players. Time series data naturally lends itself to sequence pattern mining. Zhou et al. [20] applied clustering to time series data to explore monetisation and payment behaviour of players based on a large dataset from a MMORPG game.

While most work uses data from analytics systems, some look at alternatives, such as network messages. Lee et al. [21] used server commands within World of Warcraft to receive information on other players active on the server. However, the data which can be obtained is limited to functionality provided by the server, and there are opportunities and advantages for direct integration to the game client or server, capturing all communication, most of which would be unavailable to an end user.

While the repurposing of network messages increases the amount of processing required, it creates an alternative source of data that may be more data rich, relevant or viable in situations where games do not have the required analytics data capture, as may be the case in live or legacy games. Also, as these network messages allow for the construction of a game state in real time, there are potential uses within a real-time AI pipeline, in areas such as automation and director AI systems [22,23].

With a focus on testing methods within a commercial company, Canossa et al. [24] utilised sequence pattern mining when modelling player behaviour in the game Tom Clancy's The Division, an online role playing action game with single and multiplayer elements. This was used to identify opportunities for increasing player engagement. Throughout the process, expert domain knowledge from the development team was used.

Game production environments also vary, ranging from small and micro independent teams with under 20 staff through to large AAA developers and publishers with hundreds of staff and a range of supporting departments. Maturity in the various data related aspects of game development, such as data science or user research, also varies [25].

However, there is great variation in the games available on the market, with several factors to consider, such as multiplayer vs. single player games, or competitive vs. cooperative games. The focus of analysis also varies, from looking at player behaviour over weeks or months to low level player actions within a five-minute match. For example, would this approach extend to Tower Defence games which are known to be complex, with large number of possible actions and game states [26]. Tower Defence games are a popular genre of game where players typically build defences to defeat waves of enemies. The goal is to survive for as long as possible, as the strength of the enemies increases over time. Strategic elements include the selection of defensive towers to cope with the range of enemy types, and placement of the defensive towers to optimise efficiency. Popular games in the genre include Plants vs. Zombies and Kingdom Rush.

2. Case study - Bloons

The research data in this study was obtained as part of an ongoing academic–industry collaboration to identify if player behaviour and tactics can be elicited from non-aggregated player data.

Bloons TD Battles is a player vs. player tower defence game developed by Ninja Kiwi as part of its Bloons game series [27]. It is available

Fig. 1. Bloons TD Battles gameplay.

for mobile and PC platforms. Originally released in 2012, as of late 2021, the game has 230,000 daily active players, with 400,000 matches being played daily. Ninja Kiwi has around 75 staff across two offices in New Zealand and Scotland.

Gameplay is similar to standard single player tower defence games, with players building towers, often themed around monkeys, to defeat waves of enemies, known as Bloons, before they reach the end of a track. Action takes place on one of 23 maps, each with its own layout, tracks and terrain. An example of gameplay is shown in Fig. 1.

Each buildable tower has a different set of characteristics, such as the Buccaneer Monkey which can attack from water locations, the Sniper Monkey which has a powerful long range but slow attack, or the Glue Monkey which can slow down the enemy Bloons. Bloons have a variety of characteristics adding to their challenge, such as being immune to certain damage types, regenerating their health, being undetectable to some towers, and moving at faster speeds. Defeating Bloons provides income, which can be used to build and upgrade towers.

As well as placing towers, players have a range of other options at their disposal, including boosts, which are time limited increases in defensive or attacking capability, and tower abilities, which are tower specific abilities available for defensive towers. Ahead of each match, players select their loadout, the four towers and any powers which will be available to them during the match.

The game is competitive multiplayer, with two players competing to survive the longest as they face Bloon waves of increasing difficulty. The player who runs out of lives first loses. Gameplay includes some multiplayer specific features, such as the ability to send enemy Bloon waves against the opponent, player guilds and leaderboards.

3. Method

This study integrates game development expertise into the data science process for player classification. Expert domain knowledge is important when making decisions throughout the process to reduce complexity. When manual selections are being made, such as selecting the player actions to include in the analysis, we utilise development team knowledge. Where there is a need to reduce the number of results to ensure analysis feasibility, such as between the sequence pattern mining phase and the clustering phase, information theory [28], which can rank the value of the information contained within results data, is used alongside development team knowledge.

The high-level steps of the method employed include: (a) Data Sourcing from game network messages, (b) Pre Processing, (c) Player Action Group definition from relevant player actions (d) Construction of Player Action Sequences linked to play styles and tactics, (e) Sequence Pattern Mining, (f) Sequence Scoring, (g) Sequence Labelling and Selection, (h) Cluster Analysis and Labelling. Each step will be described in more detail below.

An overview of the analysis flow is shown in Fig. 2.

3.1. Data sourcing

To obtain the data used in the analysis, we collect a sample of player data from the live game servers over a period of one week. Match replay files were identified as the source most likely to include the data suitable for identification of player tactics and 11,517 replay files were obtained from the game servers in November 2020. The match replay file is a store of all network messages sent between the clients and server during a match. This covers events such as match setup (map selection, player information), player actions (such as building towers, using powers), and game network events (such as syncing, when the game is won, or players leaving).

The replay data files containing the network messages were stored in an encoded binary format. After some header information, including the match ID, the file is structured as a series of events. Each event type has a specified format, allowing the contents of the replay file to be parsed into a readable format, such as JSON or CSV flat file.

The player data used was collected under the terms of the Ninja Kiwi data policy and provided to the researchers, with no identifiable data present or used in the analysis. No data from players under the age of eighteen was used in the analysis.

Other data, such as player win and loss stats are also included, allowing for identification of player experience and skill. As the network events contained a timestamp, this would satisfy the need to analyse player data by the order of their creation.

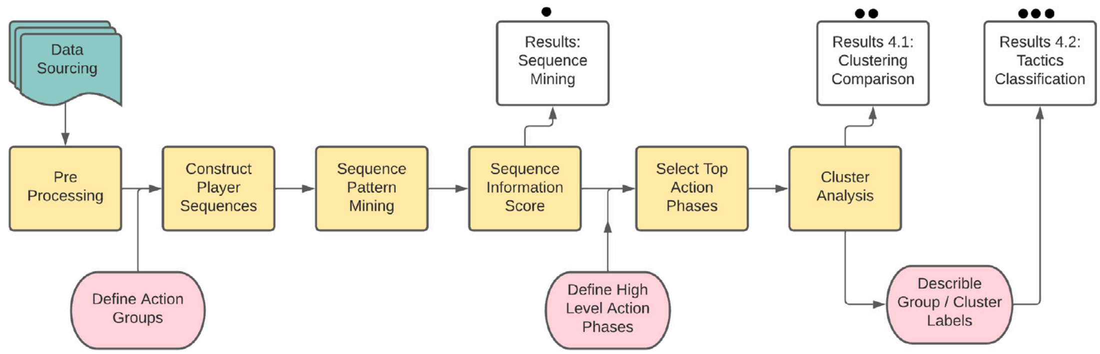

Fig. 2. Framework to determine player tactics for tower defence games. Pink rounded boxes are primary points at which domain and development team knowledge was utilised. White boxes are Results focussed used to guide the Action Phases (3.7) and generate the clustering and tactic classification (4.1, 4.2).

Table 1
Summary table for number of matches per player.

Sample dataset matches per player		
Mean	Median	Range
5.2	1.0	1.0, 162.0

Table 2
Summary table for total matches previously completed by the player.

Total matches completed by player		
Mean	Median	Range
4062	2942	0, 33,646

3.2. Data pre-processing

The dataset was cleaned before analysis. Files containing errors or no data were removed. Four matches were terminated due to automatic identification of cheating. These were excluded from the analysis. Each match has a unique identifier, and each player has a unique identifier. 237 matches involved a player with a missing unique identifier, and they were excluded from the analysis along with any duplicate matches.

Some matches had a very short duration, such as when a player won very quickly due to their opponent being idle or choosing not to compete. Very short matches were not of interest for the analysis, as they had limited scope for identifying tactics or strategies. 428 short matches, identified by having fewer than 20 player action events (such as a player building a tower or sending Bloon waves), or where the number of action events was less than 15% of all match events, were removed from the analysis.

In total, 1266 matches were removed from the analysis, leaving 9665 matches to be used in the analysis stage. As each valid match contained two players, this resulted in 19,330 examples of player behaviour. 3576 unique players were present in the replay files. These covered a range of player experience and skill (as seen by win %'s, Tables 1 and 2). A data flow diagram for the Pre-Processing step can be found in Appendix.

3.2.1. Events and actions

The match replay files contained 65 types of network event messages, many of which were not relevant for analysis of player behaviour. These included messages for setting up the match, client–server sync messages, chat messages and spectator interactions. After discussion with the development team, 14 message types were identified as being relevant for analysis. These covered the events for player actions, game status and setup information. Player actions included building towers, upgrading towers, selling towers, using abilities, using powers and sending Bloons to the opposing player. Game status events covered the game result message, while setup Information events covered sending player information and their selected loadouts. With the exception of the Bloon sending, these actions are common to tower defence games.

Each network event message contained information relevant to that event. For example, the OpponentBuiltATower event contains data on which player built the tower, which tower type was built, where it was built, and when it was built. The data for each of the 14 network event messages was transformed into a flat table structure suitable for analysis, with features as columns and rows as observations of player action.

A processing step was required to remove invalid data from the replay files. Due to a bug in the logging of events for selling towers, the replay files contained erroneous events which stated a tower has been sold, even though no corresponding tower had been built. These erroneous events were removed from the analysis.

The collected data was then analysed to identify groups of players with distinct tactics and strategies.

3.3. Defining Player Action Groups

Although the number of possible events was reduced with the selection of a subset of event types, further reduction and abstraction was required to produce a dataset which would be feasible for analysis.

Although there were only 14 event types, each one could refer to a number of distinct player actions. For example, the event for sending Bloon waves to the opponent detailed which of the 25 different Bloon types were sent by the player, and the event for upgrading a tower detailed which of the eight possible unique upgrades, spread over two separate paths, was purchased. The event for building a tower detailed which of the 33 towers was built, as well as where on the map the tower was built, given by a map coordinate. As such, the number of specific user actions across all 14 event types was high, with over 6000 specific player actions before factors such as tower position was considered.

Through discussion with the game development team, player actions were placed into high level groups, Player Action Groups. For example, the 33 individual tower types were gathered into five groups which described the function or nature of the tower. These were Attack Towers, Resource Towers, Support Towers, Track Towers and Cobra Tower. Similar groupings were created for Bloon types, grouping Bloons by their characteristics, such as cost, strength, or abilities. A unique ID was assigned to each Player Action Group. This reduced the number of player actions to 48.

3.4. Construction of user sequences

The order of actions was a known aspect of certain tactics used by the player base, such as sending the opponent one type of Bloon followed by another, or the selling of multiple towers to generate cash before sending waves of Bloons against the opponent.

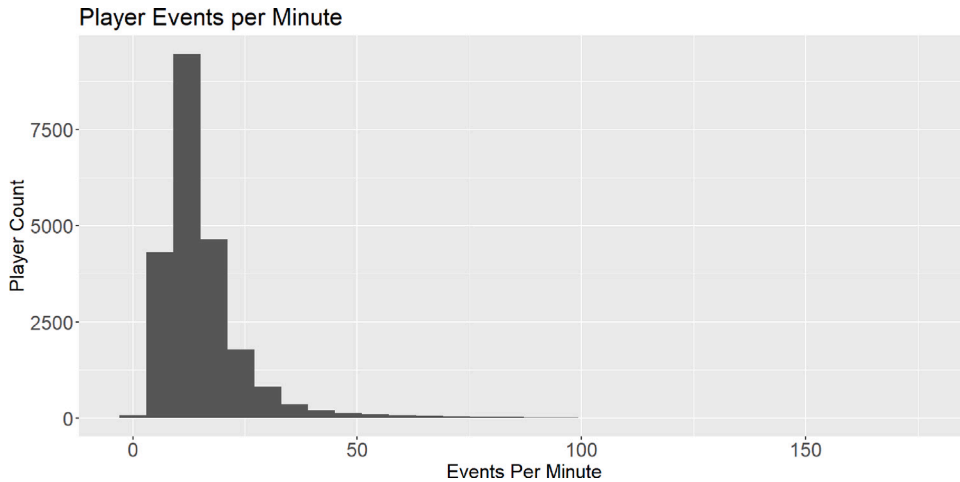

Fig. 3. Histogram of player events per minute.

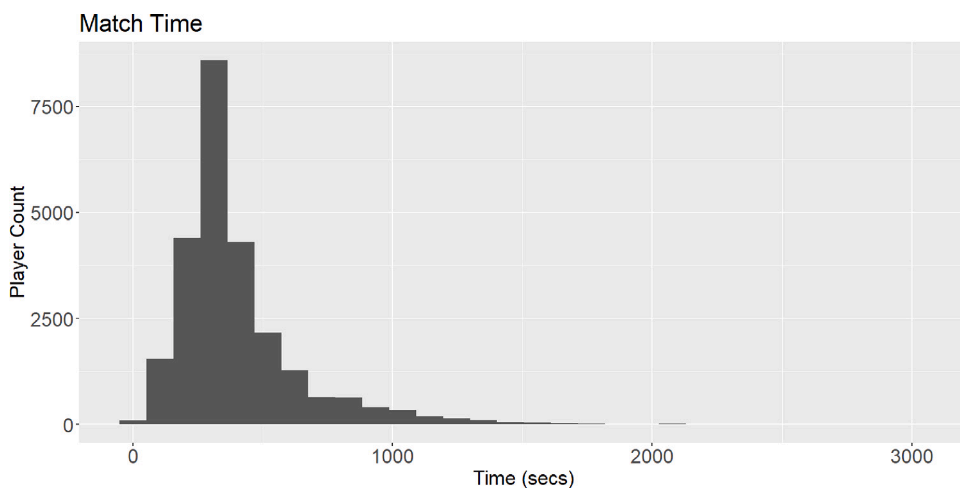

Fig. 4. Histogram of match time.

Using the Player Action Group ID, match ID, user ID and event time, a Player Action Sequence was created for each player in each match, with events ordered by time.

In total, 19,330 sequences of user actions were obtained from the valid matches. While the unique player identifier allowed for identification of all the action sequences belonging to a single player, as the objective was to identify tactics and strategies, analysis was performed at the individual action sequence level. While it would be possible for a single player to play ten matches and use three different tactics, the important factor for this analysis was the tactic, not the player.

3.5. Sequence Pattern Mining

To establish the player tactics groups, as the ordering of the actions is important, we first use sequence analysis to find common sequences of player actions within the data. After the identification of common sequences of player actions and the count of each for all players, cluster analysis will identify groups of players according to the prevalence of each sequence. The R package 'arulesSequences' was used to identify commonly occurring sequences from within the data. This package uses an implementation of the SPADE algorithm [29], an efficient and flexible algorithm with parameter customisation options.

The parameters that must be manually defined include the minimum and maximum sequence length and the maximum size of any gaps in the sequence. For this analysis, a minimum sequence length of two, maximum sequence length of ten, and no gaps were specified. During the matches, players are not constantly making actions. A sequence length of ten actions would, on average, cover time periods up to 38.0 s of a match, while considering the considerable variation between users. As shown in Figs. 3 and 4, match times are short, typically under ten minutes, meaning a 38 s long period is a significant proportion of a match. Again, domain knowledge was used as a guide for these choices.

The minimum support parameter, the frequency of occurrence within the data, can also be specified. A value of 0.05 was chosen meaning only patterns which appeared in at least 5% of all sequences would be reported. This manual value was selected with the view of creating groups that were not too niche or extreme, as would suit the intent for the usage of these groups by the development team. All other parameters were the library default.

The output of the algorithm is a list of sequences and their Support, Confidence and Lift values. Support refers to the occurrence of the pattern within the dataset, Confidence refers to the probability of the last in the sequence occurring in the data, and Lift is an interest measure used to indicate stronger associations.

A data flow diagram for the Sequence Mining stage can be found in Appendix.

3.6. Sequence Information score

Some of the mined sequences found were very common. These sequences, such as 'Build an Attack Tower' and then 'Upgrade an Attack

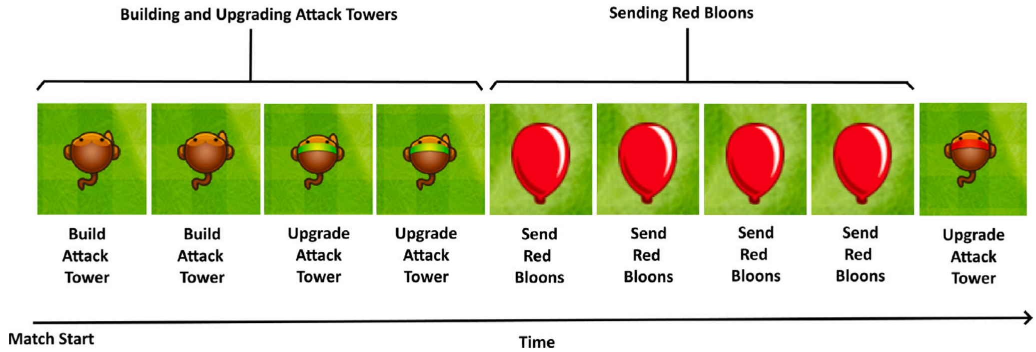

Fig. 5. Example Action Phases. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Selected action phases.

Code	Description	Code	Description
CT	Building and Upgrading Cobra Towers	TP	Using Track Power while Building Towers
TAS	Using Attack Tower Ability, and then Selling Towers	BFB	Sending MOAB, BFB or ZOMG Bloons
BWG	Sending Black or White Grouped Bloons	CR	Sending Lead, Rainbow, Zebra or Ceramic Bloons with Camo or Regeneration ability
STA	Selling Attack Towers, then Building and Upgrading Attack Towers	SR	Selling Resource Towers, then Building and Upgrading Attack Towers
BUS	Building, Upgrading and Selling Attack Towers	MLC	Sending MOAB, BFB or ZOMG Bloons, followed by Lead, Rainbow, Zebra or Ceramic Bloons
LC	Sending Lead, Rainbow, Zebra or Ceramic Bloons	TB	Using Tower Boost, and Building and Upgrading Attack Towers
BWS	Sending Black or White Spaced Bloons	SAR	Selling Attack Towers, then Building and Upgrading Resource Towers

Table 4
Default parameter values for the libraries used.

Library function	Parameter	Library default
kMeans	Algorithm Ater.max Trace	Hartigan-Wong 10 False
CSPADE	Control Maxwin	Null None

Tower' would apply to most players in most games and were of limited use in determining different tactics and strategies. Information Theory has been applied to identify sequences which contained the most information [24], and was of use when determining the action sequences to take forward to later analysis stages. A value of Shannon Entropy Equation [28], was calculated for each mined sequence, using Eq. (1), where p(x) is the probability of obtaining value x. An ordered list of results was then produced.

$$S(X) = -\sum_{i=1}^{N} p(x_i) log_2(p(x_i)) \tag{1}$$

3.7. High level labels - Action Phases

Using the output from the Sequence Information stage, manual labelling was applied to the most information rich mined sequences. This applied high level descriptive labels that described the player activity or intent from each sequence of actions.

The identified sequences covered a range of Player Action Types, such as the building, upgrading and selling of towers ("Building and Upgrading Cobra Towers", "Selling Attack Towers, then Building and Upgrading Resource Towers", "Selling Attack Towers, then Building and Upgrading for Camo Bloon Detection"), use of Bloons ("Sending MOAB, BFB or ZOMG Bloons, followed by Lead, Rainbow, Zebra or Ceramic Bloons", "Economy Boost then sending Red or Green Bloons") or use of the various powers and abilities ("Using Attack Tower Ability, and then Selling Towers", "Using Tower Boost, and Building and Upgrading Attack Towers"). An example is shown in Fig. 5. This resulted in 35 high level labels, or Action Phases. Each Action Phase was given a unique ID.

3.7.1. Top Action Phases

To ensure the cluster analysis of the data was feasible, the number of Action Phases was reduced by selecting the ones most relevant to the identification of player tactics. As the results generated are sensitive to the choices made at this stage, expert domain knowledge from the game development team was used. The identified Action Phases and information ranking scores, such as shown in Table 5, were presented to the development team. After discussion, some Action Phases were merged on consideration of them referring to similar player activity. For example, some Bloon types were regarded as similar in terms of gameplay tactics and grouped. 14 Action Phases were selected to be used in the cluster analysis phase. These are shown in Table 3.

3.8. Cluster analysis

How often each Player Action Sequence contained a selected Action Phase was counted. This produced a matrix where each row was an instance of a Player Action Sequence and each column was one of the selected Action Phases. Cluster Analysis was applied to a normalised version of this matrix.

Other work in this field has found success with different clustering techniques, including kMeans [12,20], and hierarchical methods [24].

Table 5
Sample of mined sequences, support, confidence, lift and Shannon Entropy (SE) values.

Sequence	Support	Confidence	Lift	SE
9,1,9,1,9,1,9	0.050	0.971	6.437	40.296
15,15,15,15,10,10,10,10	0.063	0.687	1.077	39.591
33,33,33,33,33,33,33,33,33,29	0.055	0.214	0.493	36.37
15,15,15,15,10,10,10	0.091	0.923	1.446	35.773
1,9,1,9,1,9	0.071	0.977	6.479	34.93
9,1,9,1,9,1	0.052	0.691	4.531	34.93
15,15,15,10,10,10,10	0.073	0.640	1.003	33.512
...	...	...	...	...

Table 6
Clustering results for kMeans methods. Results shows cluster labels, descriptions and Action Phase frequency (see Table 3).

Cluster	C1	C2	C3	C4	C5	C6	C7	C8
Tactic Group	General Play	Track Powers and Bloons	Bloon Boost/ Lead Ceramic	Big Bloons	Cobra	Wealth	Tower Boosts	Track Powers
Group size	6158	3945	2865	350	618	808	1593	2993
%	31.90%	20.40%	14.80%	1.80%	3.20%	4.20%	8.20%	15.50%
CT	0.44	0.4	0.68	2.04	12.12	0.11	0.64	0.57
TAS	0.32	0.47	0.4	1.15	0.57	4.45	1.37	0.55
BW	2.77	2.92	0.82	3.01	2.73	5.97	3.48	2.49
SAT	0.27	0.4	0.29	0.68	0.32	5.47	1.12	0.43
BUS	0.39	0.48	0.51	0.96	0.35	4.14	1.09	0.64
LC	0.52	0.68	0.71	1.72	0.91	1.17	0.94	0.71
BWS	1.61	1.75	1.9	2.39	1.52	1.74	1.73	1.56
TP	0	1	0.84	0.98	0.33	1.29	1.17	2.25
BFB	0.18	0.22	0.22	0.54	0.19	1.29	0.44	0.32
CR	0	0	1.13	0.03	0	0.04	0.17	0
SR	0.22	0.32	0.42	0.52	0.01	2.4	0.76	0.49
MLC	0	0	0.04	1.16	0	0.03	0.04	0
TB	0	0	0.02	0.02	0	0.07	1.09	0
SAR	0.06	0.06	0.11	0.14	0.02	1.31	0.22	0.09

Three different clustering techniques were applied, a partitioning based technique, a density based technique, and a hierarchical technique.

For the partitioning technique, the kMeans algorithm from the R stats package was applied [9]. The primary manual parameter choice is the number of clusters. The elbow method, determining the point of diminishing returns in terms of cluster homogeneity, was utilised to find the optimal number of clusters. Eight clusters were found to be optimal. The DBSCAN algorithm [30] was used as the density based approach. There are two parameter choices EPS (radius of the epsilon neighbourhood) and minimum cluster size. EPS value of 0.20 was selected using the elbow method. For the minimum cluster size, to ensure results were not too obscure, a value equivalent to 1% of the total cases was used.

For the hierarchical clustering method, the hclust R package [31] was applied. Again, the number of clusters must be provided, with a range of values tested. For each method, all other parameters were the library default — see Table 4. A data flow diagram for the Cluster Analysis stage can be found in Appendix.

4. Results

A sample of mined sequences, their support, confidence, lift and Shannon Entropy (SE) values is shown in Table 5. Sequences of player actions, represented by Ids are shown, such as 9: Upgrade Cobra Tower and 1: Build Cobra Tower. Sequences with high SE are deemed more information rich.

4.1. Clustering comparison

The results from three clustering methods were compared. The results are presented in Tables 6–8. The density clustering method found five clusters, which was dominated by one very large cluster, and was not desirable. A sizeable number of data points were classified as noise and were not assigned to any cluster. The hierarchical clustering method also produced results where there was one very large cluster. There were also very small clusters with few data points. This was consistent across all tested cluster numbers and was undesirable.

The kMeans algorithm produced eight clusters with a more even distribution, without one dominant cluster or obscure small clusters. Utilising domain knowledge, the kMeans algorithm was selected as the method that produced the most actionable results.

4.2. Tactics classification groups

The cluster analysis identified eight groups with distinct tactics and strategies. These are listed in Tables 6, where each has been given a title to describe the primary Action Phases used by that group.

C1 is General Play. These players do not show any strong preference for any of the selected action phases, but have an all-round style. C2 is Track Powers and Bloons, who make use of tower boosts also sending Bloons against their opponent. C3 is Bloon Boosters, who often utilise the tactic of using a Bloon Boost followed by waves of attack Bloons. C4 is Big Bloons.

This group makes most use of sending large, powerful Bloons to their opponent, often followed by other Bloons like lead or ceramic Bloons that need specific upgrades to counter. This can exploit weaknesses in an opponent's defence against these specialist Bloons. C5 is Cobra. These players focus their tactics around the Cobra tower, a high end and unique tower. C6 players are Wealth. These players focus on a long term strategy to accumulate wealth, which they can then use to overwhelm their opponent. They are far more active than other players, continually building, selling towers. C7 are Tower Boosters. These players make extensive use of tower boosts. C8 is Track Powers, where there is the strongest focus on the use of Track Powers.

5. Discussion

The method presented demonstrates a pipeline for validating and discovering player tactics to facilitate game balance. The method was applied in a commercial environment, utilising domain knowledge from

Table 7
Clustering results for Density clustering.

Density clustering		
Cluster	Number	Percent
Noise	1292	6.68%
1	14,607	72.77%
2	2425	12.55%
3	1092	5.66%
4	229	1.18%
5	225	1.16%

Table 8
Clustering results for Hierarchical clustering.

Hierarchical clustering		
Cluster	Number	Percent
1	15,664	81.03%
2	2922	15.12%
3	620	3.21%
4	108	0.54%
5	19	0.10%

different departments, e.g., analytics, tech/development, QA, community management and design. This input benefited the process of abstracting aspects of a complex game with a large action space to ensure the analysis process was feasible, and informing decisions linked to the selection of appropriate Player Actions and Action Phases.

5.1. Key findings and limitations

This method provided valuable information to Ninja Kiwi's game development team, including the prevalence of certain tactics and strategies within the player base. Importantly, since the results are not aggregated it is possible to link players and game results, the win/loss rates of groups, the success of each tactic and strategy can be explored, as can the variety of tactics used by an individual player over many matches. This can be used for game balance purposes. By using data taken from the live game environment, there can be high confidence in the validity of the findings

Feedback and discussion with the game development team highlighted opportunities for improved visualisation of the data and mined player actions sequences. The sequence mining phase can produce hundreds of sequences which need to be considered for selection or merging. These are outputted from the sequence mining phase as sequences of numerical codes, as shown in Table 5. An improved method of exploring and presenting the data would have benefited the decision-making process at points where domain knowledge was needed, lessening the knowledge required to interpret the sequences.

While some high level tactics and strategies were successfully captured, some known tactics identified by the community or design team were not detected. This was primarily due to these actions or phases not being selected for the final clustering process. Despite this, there was still actionable insight perceived by the development team. A potential limitation is the representativeness of the data. Data collection involved obtaining a random sample of player data which covered all experience and skill levels. Efforts to validate the representativeness of the collected data against these or other factors identified by the development team could be incorporated into a refined methodology.

The use of network messages as a data source suitable for player classification within multiplayer games may be relevant to the wider industry in situations where some events are not logged as part of current or planned analytics capture, such as may be the case in live or legacy games. In the case of Bloons, an analytics solution for the game was in place, but it did not capture event data at the level of detail which the game design team expected would detect some differences in tactical or strategic approaches. Aggregated data showing the number of different actions taken in a match would not be able to identify these tactical plays as the ordering would be unknown.

The outlined pipeline requires implementation of several machine learning methods, as well as game domain expertise at several stages. The technical requirements to implement the machine learning methods may be a limiting factor of adoption within industry, especially smaller, indie studios where data science maturity can be limited [32]. Conversely, the need for domain expertise from a game development studio may be a limiting factor within academic research.

5.2. Findings linked to existing literature

The application of action sequence mining (ASM) to a new genre was successful. Similar to Canossa et al. [24], the density based clustering approach was unsatisfactory, classifying a lot of data points as noise, although the proportion was less in this case (7% vs. 30%). Examining the data points classified as noise, it was clear that users with a high number of actions on one or many dimensions were seen as outliers, and classified as noise. This potentially removed valuable data around interesting tactical behaviours.

Canossa et al. [24] results showed greater success with hierarchical clustering. However, applied in this case, one very large cluster was found, which was not desirable. It was also noticeable in this case that some of the identified clusters were very small (0.1% of players), which although had distinct behaviours, were far too small a group to be of use. Manual choices in both the sequence analysis and clustering stages may account for some of these differences. These are described in Sections 3.5 and 3.8 and summarised in Table 9. The maximum sequence length of 10 was set after consideration of the patterns in tactical behaviours and the associated time span. Whereas in other contexts, long sequences of user behaviours, either in terms of timespan or number of sequences, may be of interest or necessary to identify certain behaviours; when looking at low level actions of play behaviour within a tower defence context, behaviour characteristics of different tactical approaches can be seen in shorter sequences.

This work focussed on Bloons TD Battles, a two-player competitive multiplayer tower defence game, analysing low level tactical gameplay focussed player actions within a typically 5–10 min experience. Other work [24] focused on high level patterns of user experience flow with game modes within a play session of potentially hours. The process of abstracting gameplay behaviours and elements differs. Compared to a typical analysis flow for sequence pattern mining from analytics data, the use of network messages requires an additional pre-processing step to restructure the data into an analysis suitable format.

The work presented largely aligns with previously published work, as described above, which seeks to bridge game development and data science expertise to derive player tactics. This further demonstrates the viability of the presented approach within a commercial environment and the wider application of the novel methodology to new video game genres.

5.3. Findings for the wider games industry

Game developers can tailor the analysis according to their needs at the pipeline stages where domain knowledge is incorporated (see Fig. 2). While the results from this case study grouped players by their play styles and tactics for game balance purposes, there is a range of uses for the classification data produced by the method. Examples include comparing types of player interactions and cooperation in multiplayer games against design goals, or the identification of unwanted behaviour, such as cheating or disruptive behaviour [33]. The results data could also be connected to in-game systems for personalisation or adaption, such as dynamic difficulty adjustment depending on skill or performance groupings. The output data can be carried forward to additional ML pipeline stages, where the clusters could be used

Table 9
Comparison of sequence mining and clustering parameters to previously published work.

	Game genre	Sequence mining method	Number of sequences	Clusters
This paper	Multiplayer Tower Defence	SPADE (min support 0.05, min len 2, max len 10)	14	8
Canossa et al. [24]	Online, role playing action	SPADE (min support 0.01, min len 2, no max)	13	6
Kleinman et al. [16]	Multiplayer Online Battle Arena/RTS	Manual Labelling	10	na

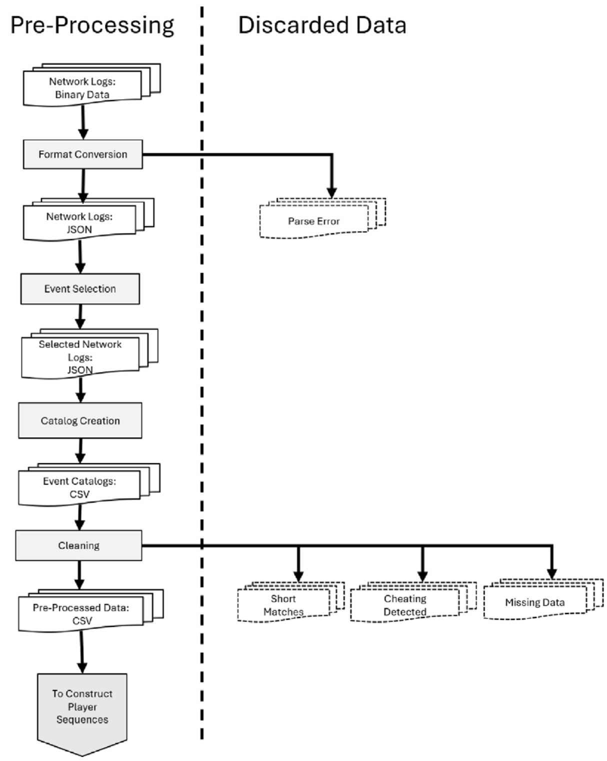

Fig. A.6. Pre processing data flow.

for the creation of automated testing agents, or for making class-based predictions of future player behaviour for gameplay, retention, or monetisation purposes [3].

The inherent complexity of many games is seen as a potential barrier to the adoption of some machine learning methods within the games industry. This case study, where the method was applied to low-level player actions in a fast-paced mobile game with a vast state space, highlights the ability of the method to be applied to such games. This is relevant to the wider gaming industry where such games are common [6].

While the machine learning skill requirement is noted as a potential limitation of the method, this case study additionally highlights the developer side benefit from industry–academic collaborations, where game developers can unlock value in stored but unanalysed data through new ways of thinking. The wider impacts of such collaborations and learnings for the wider industry are detailed by Grewar et al. [34].

6. Conclusions future work

The application of the proposed novel methodology to a rich game play dataset demonstrates three key points: (1) that expert-guided ML approaches can be used to manage the complexity of the state space associated with gameplay and hence allow previously limited approaches to be considered such as sequence mining (2) the sequence mining method together with clustering can be applied to new fast-paced genres, albeit by finetuning the parameters associated with the sequence mining and clustering methods, to elicit player tactics. (3) that fine-grained network messages prove to be a valuable data source that can be used in a commercial and live production context with avenues for future work. The challenge of visualising player actions and sequences is an active area of research [35], and could be explored, especially in the context of communication with the game development team. As the results are sensitive to decisions taken at various points during the process, potential improvements in efficiency and communication could be explored, alongside a sensitivity analysis.

There are additional benefits to using the network messages for fine-grained data collection. In addition to discovering new and validating expected tactics this rich dataset can be used to inform Quality Assurance, or the automated detection of cheating.

Ninja Kiwi are using the method presented here in an updated game development pipeline to include the modelling of player behaviours, allowing those within the development process to experiment and explore solutions. Further work could analyse the impact of this, as well as the application to other Tower Defence games in the Bloons series, each of which has variations on core tower defence features.

This work culminated in a data-driven automated testing pipeline resulting in new tools and processes for the company. This work showcases the power of ML/AI to increase productivity, lower costs and scale business processes. This proof of concept demonstrator project that has informed new data driven thinking within game development can be used as a blueprint by other games companies.

CRediT authorship contribution statement

Stuart Anderson: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Ruth Falconer:** Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Data flow diagrams

See Figs. A.6–A.8.

Data availability

The data that has been used is confidential.

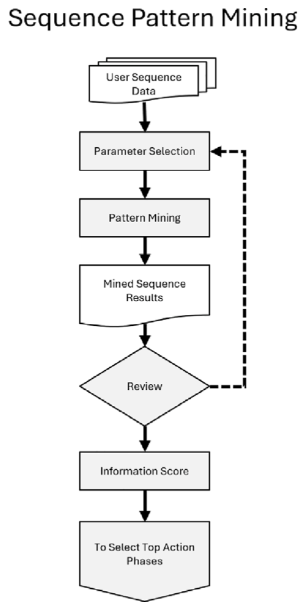

Fig. A.7. Sequence mining data flow.

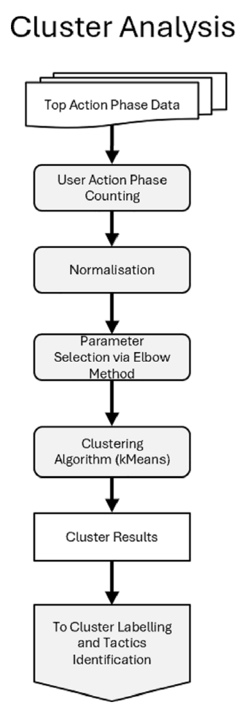

Fig. A.8. Clustering data flow.

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